b- Using the residue method, find the impulse response, h(n), of the system characterized by the following transfer function, $H(z)$:

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z-1)^2}$$

c- For the transfer function given in part (b), H(z):
(i) Plot its pole – zero diagram,
(ii) Find its region of convergence,
(iii) Discuss the stability of this system,
(iv) Find the difference equation of this system, y (n) and
(v) Draw its realization diagram using the block diagram method.

Solution

(b) To Find h(n) using Residue

$$F(z) = z^{n-1} H(z) = \frac{z^{n+1}(z+1)}{(z-0.5)(z-1)^2}$$

$$h(n) = Res_1 + Res_2$$

$$Res_1[F(z), P_1] = \frac{z^{n+1}(z+1)}{(z-1)^2}\Big|_{z=0.5}$$

$$Res_1[F(z), 0.5] = \frac{(0.5)^{n+1} * 1.5}{+0.25} = +4 * 1.5 (0.5)^{n+1} = +6 (0.5)^{n+1} = 3(0.5)^n$$

$$Res_2[F(z), 1] = \frac{d}{dz} \cdot \frac{z^{n+1}(z+1)}{(z-0.5)} = \frac{d}{dz} \cdot \frac{z^{n+2} + z^{n+1}}{(z-0.5)}$$

$$= \frac{(z-0.5)\left((n+2)z^{n+1} + (n+1)z^n\right) - \left(z^{n+2} + z^{n+1}\right)}{(z-0.5)^2}\Bigg|_{z=1}$$

$$= \frac{0.5\left((n+2)1^{n+1} + (n+1)1^n\right) - \left(1^{n+2} + 1^{n+1}\right)}{0.25} = \frac{0.5(n+2+n+1) - 2}{0.25}$$

$$= 2(2n+1) - 8 = 2(2n-1)$$

$$h(n) = 3(0.5)^n + 2(2n-1)$$

at $n=0$ $\quad h(0) = 3 + 2(-1) = 1$

at $n=1$ $\quad h(1) = 3 * 0.5 + 2(1) = 1.5 + 2 = 3.5$

at $n=2$ $\quad h(2) = 3(0.5)^2 + 2(4-1) = 0.75 + 6 = 6.75$

at $n=3$ $\quad h(3) = 3(0.5)^3 + 2(6-1) = 0.375 + 10 = 10.375$

$$h(n) = \{1, 3.5, 6.75, 10.375\}$$

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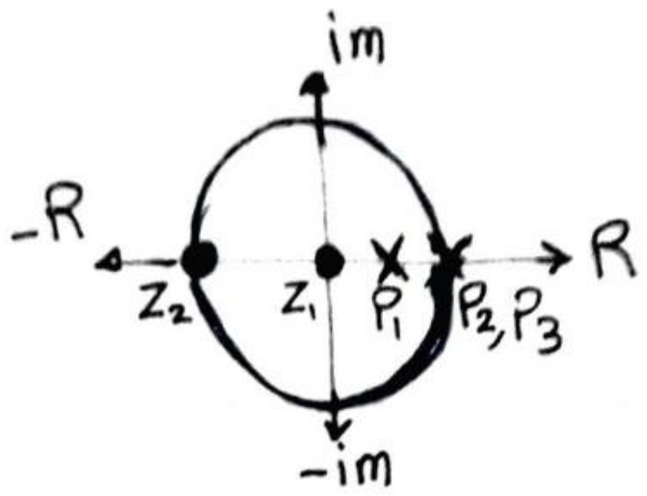
∈ For The Transfer Function

(i) Plot its Pole-Zero diagram.

$$H(z) = \frac{z^2(z+0)}{(z-0.5)(z-1)^2}$$

Zeros $z=0$ $z=-1$

Poles $P_1=0.5$ $P_2, P_3=1$



(ii) Find its region of Convergence

ROC = All $z - \{0.5, 1, \infty\}$

(iii) Discuss The stability of this system

"marigen stable"

(iv) Find The difference eQuation

$$H(z) = \frac{z^2(z+1)}{(z-0.5)(z^2-2z+1)} = \frac{z^3+z^2}{z^3-2z^2+z-0.5z^2+z-0.5}$$

$$H(z) = \frac{z^3+z^2}{z^3-2.5z^2+2z-0.5} = \frac{y(z)}{x(z)} = \frac{z^{-3}}{z-3}$$

$$\frac{y(z)}{x(z)} = \frac{1+z^{-1}}{1-2.5z^{-1}+2z^{-2}-0.5z^{-3}}$$

$$y(z)\left[1-2.5z^{-1}+2z^{-2}-0.5z^{-3}\right] = x(z)\left[1+z^{-1}\right]$$

$$y(z)-2.5y(z)z^{-1}+2y(z)z^{-2}-0.5y(z)z^{-3} = x(z)+x(z)z^{-1}$$

Take iZt For both Side

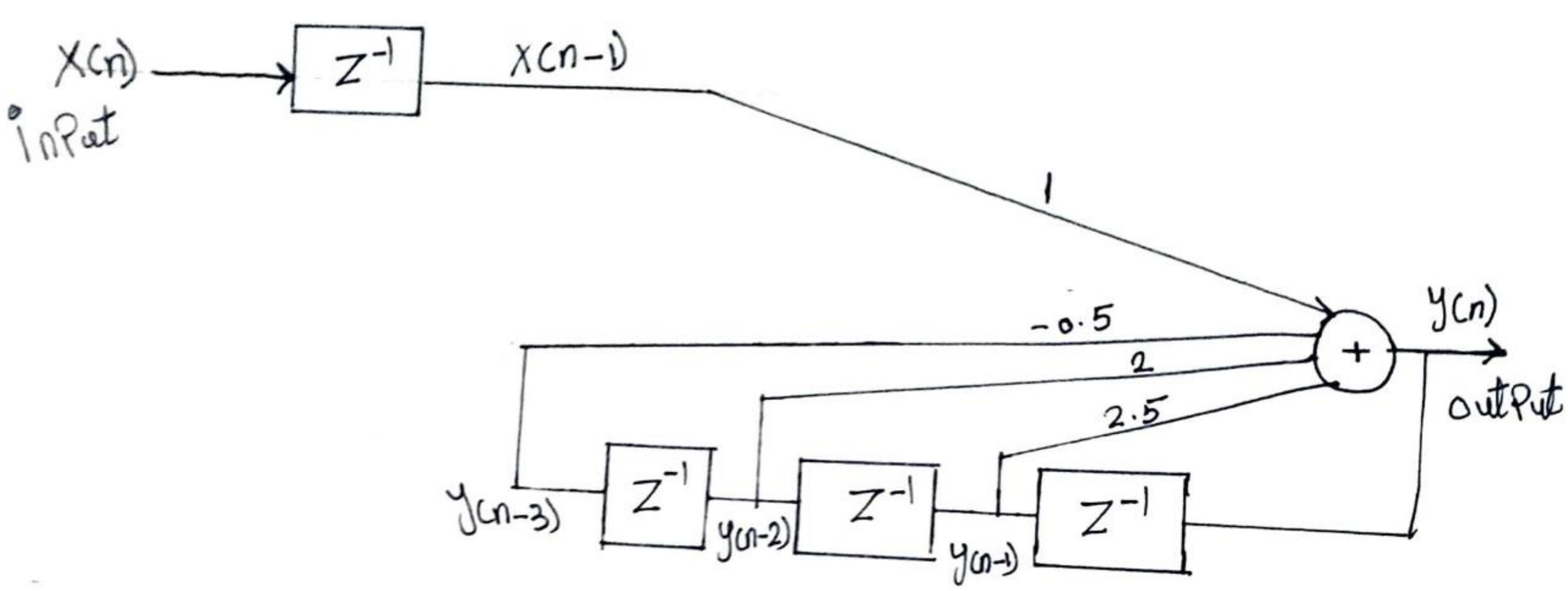
$$y(n) = x(n)+x(n-1)+2.5y(n-1)+2y(n-2)-0.5y(n-3)$$

↑ Difference eQuation

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(v) Draw its realization diagram using The block diagram method

$$y(n) = x(n) + x(n-1) + 2.5\,y(n-1) + 2y(n-2) - 0.5y(n-3)$$



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